

MATH 180 Strategies of Problem Solving

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Outline

- 1 SoPS Lecture 7
 - Battleship with Radar
 - Firing algorithms
 - Wrapping up

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Review

Last time

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- we explored Battleship with radar

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- we talked about inverse problems

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Last time

- we explored Battleship with radar
- we talked about inverse problems
- we discussed existence and uniqueness

Existence

Problem 1

Existence

Problem 1

- (a) Find a Battleship fleet with X-ray data
Rows: 4, 4, 4, 4, 1, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0

Existence

Problem 1

- (a) Find a Battleship fleet with X-ray data
Rows: 4, 4, 4, 4, 1, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0
- (b) Find a Battleship fleet with X-ray data
Rows: 0, 2, 5, 1, 4, 2, 1, 1, 1, 0 Cols: 0, 1, 2, 1, 2, 5, 1, 1, 1, 3

Non-existence

Problem 2

Non-existence

Problem 2

- (a) Explain why there is no fleet with the *X-ray* data Rows:
4, 2, 4, 3, 3, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0

Non-existence

Problem 2

- (a) Explain why there is no fleet with the *X-ray* data Rows: 4, 2, 4, 3, 3, 0, 0, 0, 0, 0 Cols: 5, 4, 4, 4, 0, 0, 0, 0, 0, 0
- (b) Explain why there is no fleet with the *X-ray* data Rows: 0, 2, 2, 5, 4, 4, 0, 0, 0, 0 Cols: 1, 2, 2, 3, 2, 2, 0, 2, 1, 2

Non-uniqueness

Problem 3

Non-uniqueness

Problem 3

- (a) Find an example of a fleet whose position is uniquely determined by its radar data.

Non-uniqueness

Problem 3

- (a) Find an example of a fleet whose position is uniquely determined by its radar data.
- (b) Find an example of a fleet whose position is “stealthy”: whose radar data actually describes several different potential fleet positions. Show the other fleets.

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Static firing algorithms

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Vocabulary

A **static firing algorithm** is one which fires in a pre-determined pattern, not changing whether each shot is a hit or a miss.

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A **static firing algorithm** is one which fires in a pre-determined pattern, not changing whether each shot is a hit or a miss.

Question

Imagine we want to explore *mostly* static firing algorithms, where if you get a hit, it searches for the ship that you hit but otherwise fires in a predetermined pattern. What is the best pattern to sink a randomly generated fleet?

Battleship simulator

Activity

Use the [Battleship Simulator \(link\)](#) to explore different firing patterns. Try to find the pattern that averages the least shots.

Battleship simulator

Activity

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Share your best pattern



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Discussion

Discussion

Static vs Dynamic

A **dynamic** firing algorithm is one where the shots change depending on what happens on the board.

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Question 1

What makes a good static firing algorithm?

Discussion

Static vs Dynamic

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Question 1

What makes a good static firing algorithm?

Question 2

Can you think of an interesting dynamic firing algorithm that might improve over the static algorithms we found?

Question of the Day

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Puzzle

Suppose there is an enemy submarine at an unknown location on the integer number line, and each minute, it moves at an unknown, fixed integer velocity. You may fire one missile per minute somewhere on the number line in an attempt to hit the submarine. Is there a strategy to guarantee you will eventually hit the submarine?

Final thoughts

Reflection

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- What was the static pattern you found that worked the best?

Final thoughts

Reflection

- What was the static pattern you found that worked the best?
- What dynamic algorithm would you suggest to improve your static algorithm.

Submit your responses by **Tonight!**